SMARTFIT Inc.

EMC TEST REPORT FOR

SMARTfit™ Pod Target
Model: 3-56228

Tested to The Following Standards:

EN 301 489-3 v2.1.1 / EN 301 489-1 v2.1.1

Report No.: 99820-8

Date of issue: June 2, 2017

We strive to create long-term, trust based relationships by providing sound, adaptive, customer first testing services. We embrace each of our customers' unique EMC challenges, not as an interruption to set processes, but rather as the reason we are in business.

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Testing the Future
CKC LABORATORIES, INC.

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ADMINISTRATIVE INFORMATION

Test Report Information

REPORT PREPARED FOR:

SMARTFIT Inc.
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Camarillo, CA 93012

Representative: Jim Manley
Customer Reference Number: 2668

REPORT PREPARED BY:

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Mariposa, CA 95338

Project Number: 99820

DATE OF EQUIPMENT RECEIPT: May 5, 2017
DATE(S) OF TESTING: May 5-24, 2017

Report Authorization

The test data contained in this report documents the observed testing parameters pertaining to and are relevant for only the sample equipment tested in the agreed upon operational mode(s) and configuration(s) as identified herein. Compliance assessment remains the client's responsibility. This report may not be used to claim product endorsement by A2LA or any government agencies. This test report has been authorized for release under quality control from CKC Laboratories, Inc.

Steve Behm
Director of Quality Assurance & Engineering Services
CKC Laboratories, Inc.

Test Facility Information



Our laboratories are configured to effectively test a wide variety of product types. CKC utilizes first class test equipment, anechoic chambers, data acquisition and information services to create accurate, repeatable and affordable test results.

TEST LOCATION(S):
CKC Laboratories, Inc.
110 Olinda Place
Brea, CA 92823

Software Versions

CKC Laboratories Proprietary Software	Version
EMITest Emissions	5.03.02
EMITest Immunity	5.03.02

Site Registration & Accreditation Information

Location	CB #	TAIWAN	CANADA	FCC	JAPAN
Brea A, CA	US0060	SL2-IN-E-1146R	3082D-1	US1025	A-0147

SUMMARY OF RESULTS

Standard / Specification: EN 301 489-3 v2.1.1 / EN 301 489-1 v2.1.1

Test Procedure	Description	Modifications	Results
EN 301 489-3	Conducted Emissions	NA	Pass
EN 301 489-3	Telecom Conducted Emissions	NA	NA
EN 301 489-3	Radiated Emissions	NA	NA
EN 61000-3-2 (2014)[1]	Harmonic Emissions	NA	Pass
EN 61000-3-3 (2013)[1]	Voltage Fluctuations and Flicker Emissions	NA	Pass
EN 61000-4-2 (2009)	Electrostatic Discharge	NA	Pass
EN 61000-4-3 (2006)+A1+A2	Radiated Immunity	NA	Pass
EN 61000-4-4 (2012)	Electrical Fast Transient Burst	NA	Pass
EN 61000-4-5 (2006)	Surge	NA	Pass
EN 61000-4-6 (2009)	Conducted Immunity	NA	Pass
EN 61000-4-11 (2004)	Voltage Dips and Interrupts	NA	Pass
ISO 7637-2 (2011)	Transients and Surges in the Vehicle Environment	NA	NA

NA= Not Applicable

[1]The AC mains current harmonics and voltage flicker test requirements are derived from European Union CE mark directives.

Modifications During Testing

This list is a summary of the modifications made to the equipment during testing.

Summary of Conditions
No modifications were made during testing.

Modifications listed above must be incorporated into all production units.

Conditions During Testing

This list is a summary of the conditions noted to the equipment during testing.

Summary of Conditions
Testing was performed to EN 301 489-3 v2.1.1 which at the time of testing was not an approved standard in Europe.

EQUIPMENT UNDER TEST

During testing numerous configurations may have been utilized. The configurations listed below support compliance to the standard(s) listed in the Summary of Results section.

Configuration 1 and Configuration A

Equipment Tested:

Device	Manufacturer	Model #	S/N
SMARTfit™ Pod Target	SMARTFIT Inc.	3-56228	NA
Power Supply	Triad Magnetics	WSU060-3000	NA

Support Equipment:

Device	Manufacturer	Model #	S/N
SMARTfit™ CPU Controller	SMARTFIT Inc.	3-56322	NA

EN 301 489-3

Conducted Emissions

Test Notes: Conducted Disturbances at Mains Terminals, LISN method.

Test Setup / Conditions / Data

Test Location:	CKC Laboratories Inc. • 110 N. Olinda Place • Brea, CA 92823 • 714-993-6112		
Customer:	**SMARTFIT Inc.**		
Specification:	**EN 301 489-3 AC/DC Mains Input/Output - Average**		
Work Order #:	**99820**	Date:	5/5/2017
Test Type:	**Conducted Emissions**	Time:	15:25:24
Tested By:	Don Nguyen	Sequence#:	3
Software:	EMITest 5.03.02		230V 50Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 1			

Support Equipment:

Device	Manufacturer	Model #	S/N
Configuration 1			

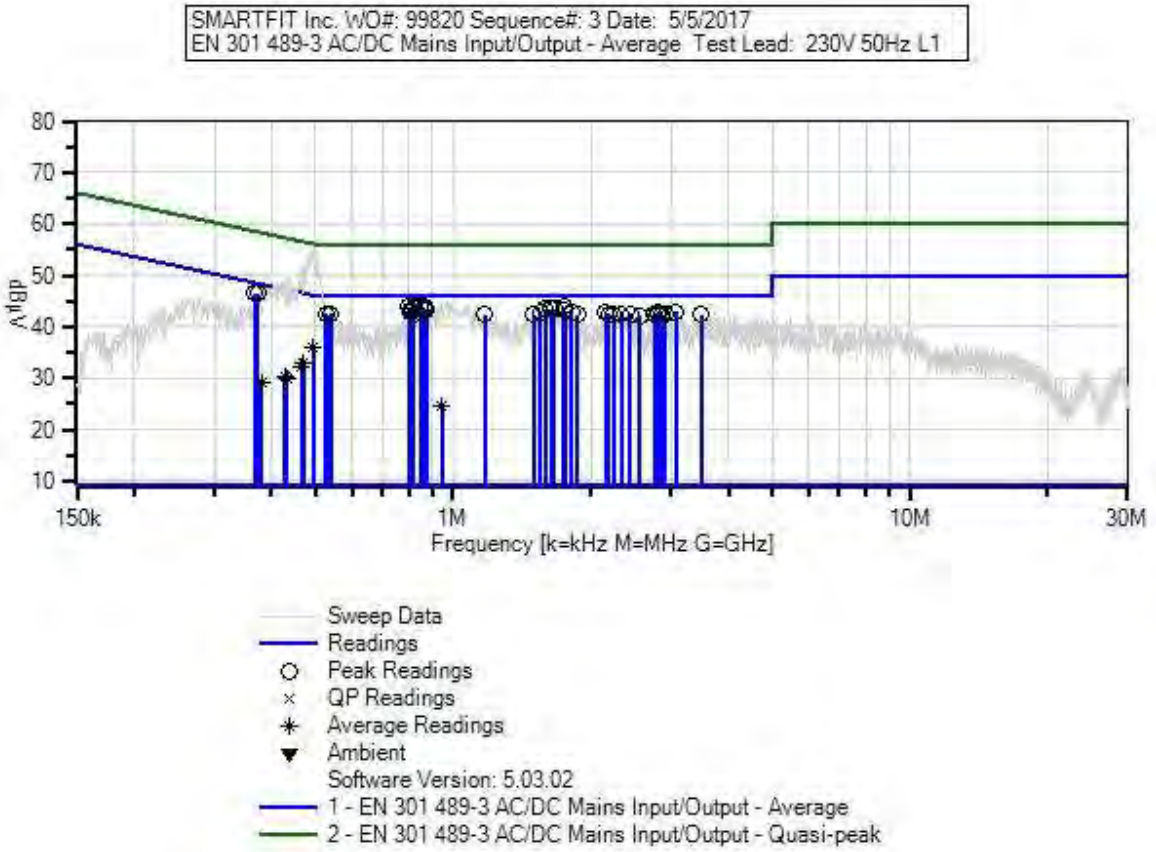
Test Conditions / Notes:

The EUT is placed on tabletop. The EUT is connected to 6Vdc adapter. The input voltage to ACDC adapter is 230Vac/50Hz. A 3.7Vdc lithium battery is installed inside the EUT.
The EUT is tested in preferred orientation declared by the manufacturer.
The EUT is set to continuously transmit.

Operating frequency: 434MHz
Antenna gain: 0dBi
Software output power: <1mW

Frequency range of measurement = 150kHz-30MHz
150kHz-30MHz, RBW=9kHz, VBW=9kHz

Temperature: 26°C, Humidity: 40%, Pressure: 100.1kPa.
Site A.
Test Method: EN 55032 (2015) annex A.3


SMARTFIT Inc. WO#: 99820 Sequence#: 3 Date: 5/5/2017
EN 301 489-3 AC/DC Mains Input/Output - Average Test Lead: 230V 50Hz L1
80
70
60
50
40
30
20
10
dBµV
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Readings
Peak Readings
QP Readings
Average Readings
Ambient
Software Version: 5.03.02
1 - EN 301 489-3 AC/DC Mains Input/Output - Average
2 - EN 301 489-3 AC/DC Mains Input/Output - Quasi-peak

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
T1	ANC00079	Attenuator		2/3/2017	2/3/2019
T2	ANP04358	Cable	RG142	3/14/2016	3/14/2018
T3	AN00969A	50uH LISN-Line 1 (L1)	3816/2NM	3/14/2017	3/14/2019
	AN00969A	50uH LISN-Line2 (L2)	3816/2NM	3/14/2017	3/14/2019
	AN02672	Spectrum Analyzer	E4446A	3/2/2017	3/2/2019
T4	AN02610	High Pass Filter	HE9615-150K-50-720B	10/22/2015	10/22/2017

Measurement Data: Reading listed by margin. Test Lead: L1

#	Freq MHz	Rdng dBµV	T1 dB	T2 dB	T3 dB	T4 dB	Dist Table	Corr dBµV	Spec dBµV	Margin dB	Polar Ant
1	373.979k	36.3	+9.8	+0.1	+0.0	+0.2	+0.0	46.4	48.4	-2.0	L1
2	821.211k	33.9	+9.8	+0.1	+0.0	+0.2	+0.0	44.0	46.0	-2.0	L1
3	848.117k	33.8	+9.8	+0.1	+0.0	+0.2	+0.0	43.9	46.0	-2.1	L1
4	800.122k	33.7	+9.8	+0.1	+0.0	+0.2	+0.0	43.8	46.0	-2.2	L1
5	367.434k	36.3	+9.8	+0.1	+0.0	+0.2	+0.0	46.4	48.6	-2.2	L1
6	1.745M	33.8	+9.8	+0.1	+0.0	+0.1	+0.0	43.8	46.0	-2.2	L1
7	859.025k	33.7	+9.8	+0.1	+0.0	+0.2	+0.0	43.8	46.0	-2.2	L1
8	875.024k	33.4	+9.8	+0.1	+0.0	+0.2	+0.0	43.5	46.0	-2.5	L1
9	864.843k	33.4	+9.8	+0.1	+0.0	+0.2	+0.0	43.5	46.0	-2.5	L1
10	1.660M	33.5	+9.8	+0.1	+0.0	+0.1	+0.0	43.5	46.0	-2.5	L1
11	1.592M	33.5	+9.8	+0.1	+0.0	+0.1	+0.0	43.5	46.0	-2.5	L1
12	1.643M	33.4	+9.8	+0.1	+0.0	+0.1	+0.0	43.4	46.0	-2.6	L1
13	805.939k	33.2	+9.8	+0.1	+0.0	+0.2	+0.0	43.3	46.0	-2.7	L1
14	869.206k	33.2	+9.8	+0.1	+0.0	+0.2	+0.0	43.3	46.0	-2.7	L1
15	815.393k	33.1	+9.8	+0.1	+0.0	+0.2	+0.0	43.2	46.0	-2.8	L1
16	867.752k	33.1	+9.8	+0.1	+0.0	+0.2	+0.0	43.2	46.0	-2.8	L1
17	873.569k	33.1	+9.8	+0.1	+0.0	+0.2	+0.0	43.2	46.0	-2.8	L1
18	1.753M	33.0	+9.8	+0.1	+0.0	+0.1	+0.0	43.0	46.0	-3.0	L1

19	3.067M	32.7	+9.8	+0.2	+0.1	+0.1	+0.0	42.9	46.0	-3.1	L1
20	2.821M	32.8	+9.8	+0.2	+0.0	+0.1	+0.0	42.9	46.0	-3.1	L1
21	803.758k	32.8	+9.8	+0.1	+0.0	+0.2	+0.0	42.9	46.0	-3.1	L1
22	1.553M	32.9	+9.8	+0.1	+0.0	+0.1	+0.0	42.9	46.0	-3.1	L1
23	1.821M	32.7	+9.8	+0.1	+0.0	+0.1	+0.0	42.7	46.0	-3.3	L1
24	2.153M	32.7	+9.8	+0.1	+0.0	+0.1	+0.0	42.7	46.0	-3.3	L1
25	525.965k	32.5	+9.8	+0.1	+0.0	+0.2	+0.0	42.6	46.0	-3.4	L1
26	2.804M	32.5	+9.8	+0.2	+0.0	+0.1	+0.0	42.6	46.0	-3.4	L1
27	819.029k	32.4	+9.8	+0.1	+0.0	+0.2	+0.0	42.5	46.0	-3.5	L1
28	540.509k	32.4	+9.8	+0.1	+0.0	+0.2	+0.0	42.5	46.0	-3.5	L1
29	2.187M	32.5	+9.8	+0.1	+0.0	+0.1	+0.0	42.5	46.0	-3.5	L1
30	2.438M	32.4	+9.8	+0.2	+0.0	+0.1	+0.0	42.5	46.0	-3.5	L1
31	2.336M	32.4	+9.8	+0.2	+0.0	+0.1	+0.0	42.5	46.0	-3.5	L1
32	1.502M	32.4	+9.8	+0.1	+0.0	+0.1	+0.0	42.4	46.0	-3.6	L1
33	1.171M	32.3	+9.8	+0.1	+0.0	+0.2	+0.0	42.4	46.0	-3.6	L1
34	536.146k	32.3	+9.8	+0.1	+0.0	+0.2	+0.0	42.4	46.0	-3.6	L1
35	3.501M	32.2	+9.8	+0.2	+0.1	+0.1	+0.0	42.4	46.0	-3.6	L1
36	2.757M	32.3	+9.8	+0.2	+0.0	+0.1	+0.0	42.4	46.0	-3.6	L1
37	1.872M	32.4	+9.8	+0.1	+0.0	+0.1	+0.0	42.4	46.0	-3.6	L1
38	2.247M	32.2	+9.8	+0.2	+0.0	+0.1	+0.0	42.3	46.0	-3.7	L1
39	2.855M	32.2	+9.8	+0.2	+0.0	+0.1	+0.0	42.3	46.0	-3.7	L1
40	2.910M	32.2	+9.8	+0.2	+0.0	+0.1	+0.0	42.3	46.0	-3.7	L1
41	2.566M	32.1	+9.8	+0.2	+0.0	+0.1	+0.0	42.2	46.0	-3.8	L1
42	494.920k	26.0	+9.8	+0.1	+0.0	+0.2	+0.0	36.1	46.1	-10.0	L1
Ave											
^	494.920k	44.8	+9.8	+0.1	+0.0	+0.2	+0.0	54.9	46.1	+8.8	L1
^	495.422k	44.8	+9.8	+0.1	+0.0	+0.2	+0.0	54.9	46.1	+8.8	L1

45 Ave	471.425k	22.9	+9.8	+0.1	+0.0	+0.2	+0.0	33.0	46.5	-13.5	L1
^	471.425k	41.9	+9.8	+0.1	+0.0	+0.2	+0.0	52.0	46.5	+5.5	L1
47 Ave	465.607k	21.9	+9.8	+0.1	+0.0	+0.2	+0.0	32.0	46.6	-14.6	L1
^	465.607k	41.1	+9.8	+0.1	+0.0	+0.2	+0.0	51.2	46.6	+4.6	L1
49 Ave	427.793k	20.1	+9.8	+0.1	+0.0	+0.2	+0.0	30.2	47.3	-17.1	L1
^	427.792k	39.2	+9.8	+0.1	+0.0	+0.2	+0.0	49.3	47.3	+2.0	L1
51 Ave	433.610k	19.8	+9.8	+0.1	+0.0	+0.2	+0.0	29.9	47.2	-17.3	L1
^	433.610k	39.3	+9.8	+0.1	+0.0	+0.2	+0.0	49.4	47.2	+2.2	L1
^	435.064k	39.1	+9.8	+0.1	+0.0	+0.2	+0.0	49.2	47.2	+2.0	L1
^	430.701k	39.0	+9.8	+0.1	+0.0	+0.2	+0.0	49.1	47.2	+1.9	L1
55 Ave	380.524k	19.1	+9.8	+0.1	+0.0	+0.2	+0.0	29.2	48.3	-19.1	L1
^	380.524k	37.1	+9.8	+0.1	+0.0	+0.2	+0.0	47.2	48.3	-1.1	L1
57 Ave	945.249k	14.6	+9.8	+0.1	+0.0	+0.2	+0.0	24.7	46.0	-21.3	L1
^	945.248k	34.9	+9.8	+0.1	+0.0	+0.2	+0.0	45.0	46.0	-1.0	L1

Test Location:	CKC Laboratories Inc. • 110 N. Olinda Place • Brea, CA 92823 • 714-993-6112		
Customer:	**SMARTFIT Inc.**		
Specification:	**EN 301 489-3 AC/DC Mains Input/Output - Average**		
Work Order #:	**99820**	Date:	5/5/2017
Test Type:	**Conducted Emissions**	Time:	15:28:25
Tested By:	Don Nguyen	Sequence#:	4
Software:	EMITest 5.03.02		230V 50Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 1			

Support Equipment:

Device	Manufacturer	Model #	S/N
Configuration 1			

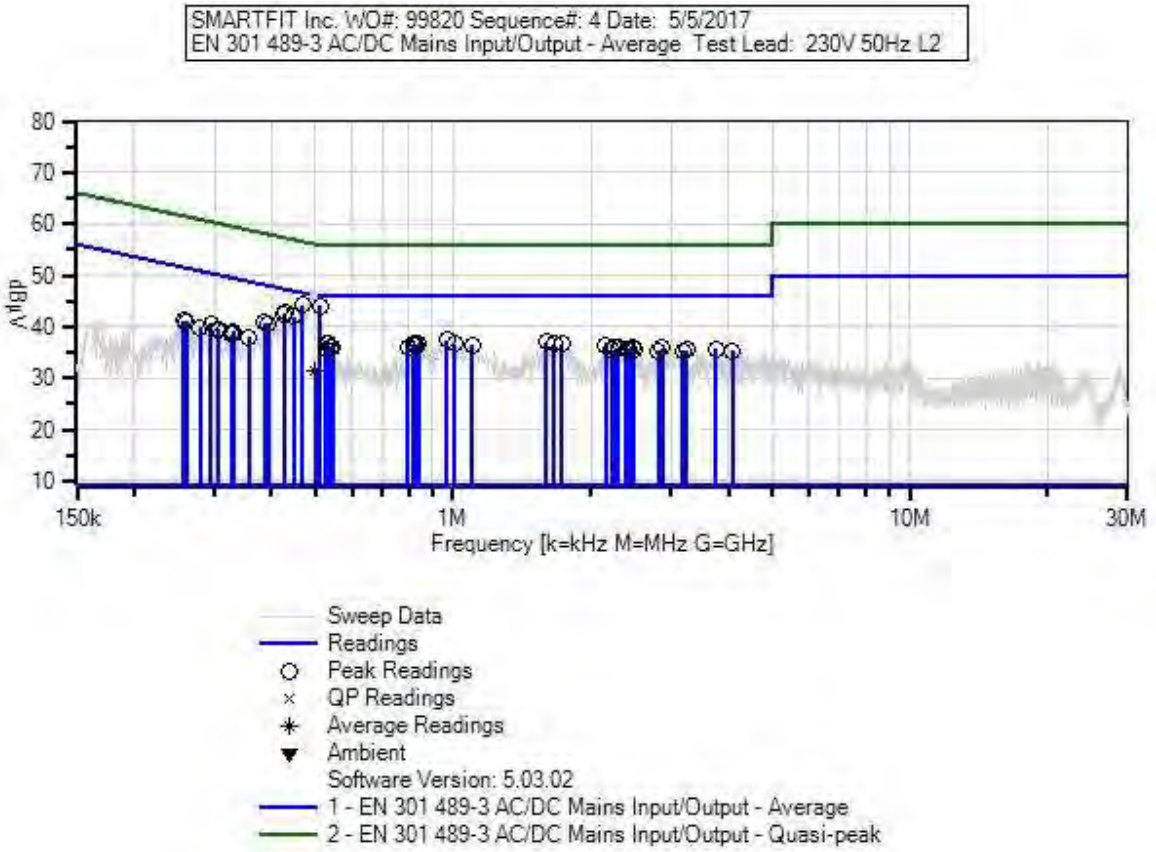
Test Conditions / Notes:

The EUT is placed on tabletop. The EUT is connected to 6Vdc adapter. The input voltage to ACDC adapter is 230Vac/50Hz. A 3.7Vdc lithium battery is installed inside the EUT.
The EUT is tested in preferred orientation declared by the manufacturer.
The EUT is set to continuously transmit.

Operating frequency: 434MHz
Antenna gain: 0dBi
Software output power: <1mW

Frequency range of measurement = 150kHz-30MHz
150kHz-30MHz, RBW=9kHz, VBW=9kHz

Temperature: 26°C, Humidity: 40%, Pressure: 100.1kPa.
Site A.
Test Method: EN 55032 (2015) annex A.3


SMARTFIT Inc. WO#: 99820 Sequence#: 4 Date: 5/5/2017
EN 301 489-3 AC/DC Mains Input/Output - Average Test Lead: 230V 50Hz L2
80
70
60
50
40
30
20
10
dBµV
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Readings
Peak Readings
QP Readings
Average Readings
Ambient
Software Version: 5.03.02
1 - EN 301 489-3 AC/DC Mains Input/Output - Average
2 - EN 301 489-3 AC/DC Mains Input/Output - Quasi-peak

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
T1	ANC00079	Attenuator		2/3/2017	2/3/2019
T2	ANP04358	Cable	RG142	3/14/2016	3/14/2018
	AN00969A	50uH LISN-Line 1 (L1)	3816/2NM	3/14/2017	3/14/2019
T3	AN00969A	50uH LISN-Line2 (L2)	3816/2NM	3/14/2017	3/14/2019
	AN02672	Spectrum Analyzer	E4446A	3/2/2017	3/2/2019
T4	AN02610	High Pass Filter	HE9615-150K-50-720B	10/22/2015	10/22/2017

Measurement Data: Reading listed by margin. Test Lead: L2

#	Freq MHz	Rdng dBµV	T1 dB	T2 dB	T3 dB	T4 dB	Dist Table	Corr dBµV	Spec dBµV	Margin dB	Polar Ant
1	510.694k	33.9	+9.8	+0.1	+0.0	+0.2	+0.0	44.0	46.0	-2.0	L2
2	468.516k	34.2	+9.8	+0.1	+0.0	+0.2	+0.0	44.3	46.5	-2.2	L2
3	429.247k	32.6	+9.8	+0.1	+0.0	+0.2	+0.0	42.7	47.3	-4.6	L2
4	426.338k	32.5	+9.8	+0.1	+0.0	+0.2	+0.0	42.6	47.3	-4.7	L2
5	448.154k	31.8	+9.8	+0.1	+0.0	+0.2	+0.0	41.9	46.9	-5.0	L2
6	387.069k	30.7	+9.8	+0.1	+0.0	+0.2	+0.0	40.8	48.1	-7.3	L2
7	393.614k	30.5	+9.8	+0.1	+0.0	+0.2	+0.0	40.6	48.0	-7.4	L2
8	966.512k	27.3	+9.8	+0.1	+0.0	+0.2	+0.0	37.4	46.0	-8.6	L2
9	1.600M	27.0	+9.8	+0.1	+0.0	+0.1	+0.0	37.0	46.0	-9.0	L2
10	531.055k	26.8	+9.8	+0.1	+0.0	+0.2	+0.0	36.9	46.0	-9.1	L2
11	837.936k	26.8	+9.8	+0.1	+0.0	+0.2	+0.0	36.9	46.0	-9.1	L2
12	822.665k	26.6	+9.8	+0.1	+0.0	+0.2	+0.0	36.7	46.0	-9.3	L2
13	820.483k	26.6	+9.8	+0.1	+0.0	+0.2	+0.0	36.7	46.0	-9.3	L2
14	525.965k	26.6	+9.8	+0.1	+0.0	+0.2	+0.0	36.7	46.0	-9.3	L2
15	1.728M	26.7	+9.8	+0.1	+0.0	+0.1	+0.0	36.7	46.0	-9.3	L2
16	1.660M	26.7	+9.8	+0.1	+0.0	+0.1	+0.0	36.7	46.0	-9.3	L2
17	1.009M	26.6	+9.8	+0.1	+0.0	+0.2	+0.0	36.7	46.0	-9.3	L2
18	831.392k	26.4	+9.8	+0.1	+0.0	+0.2	+0.0	36.5	46.0	-9.5	L2

19	1.098M	26.4	+9.8	+0.1	+0.0	+0.2	+0.0	36.5	46.0	-9.5	L2
20	2.162M	26.4	+9.8	+0.1	+0.1	+0.1	+0.0	36.5	46.0	-9.5	L2
21	824.847k	26.2	+9.8	+0.1	+0.0	+0.2	+0.0	36.3	46.0	-9.7	L2
22	295.441k	30.6	+9.8	+0.1	+0.0	+0.1	+0.0	40.6	50.4	-9.8	L2
23	536.873k	26.0	+9.8	+0.1	+0.0	+0.2	+0.0	36.1	46.0	-9.9	L2
24	2.463M	25.8	+9.8	+0.2	+0.1	+0.1	+0.0	36.0	46.0	-10.0	L2
25	2.264M	25.7	+9.8	+0.2	+0.1	+0.1	+0.0	35.9	46.0	-10.1	L2
26	795.758k	25.7	+9.8	+0.1	+0.0	+0.2	+0.0	35.8	46.0	-10.2	L2
27	544.145k	25.7	+9.8	+0.1	+0.0	+0.2	+0.0	35.8	46.0	-10.2	L2
28	2.867M	25.6	+9.8	+0.2	+0.1	+0.1	+0.0	35.8	46.0	-10.2	L2
29	2.298M	25.6	+9.8	+0.2	+0.1	+0.1	+0.0	35.8	46.0	-10.2	L2
30	256.899k	31.2	+9.8	+0.1	+0.0	+0.1	+0.0	41.2	51.5	-10.3	L2
31	2.493M	25.5	+9.8	+0.2	+0.1	+0.1	+0.0	35.7	46.0	-10.3	L2
32	3.761M	25.5	+9.8	+0.2	+0.1	+0.1	+0.0	35.7	46.0	-10.3	L2
33	330.347k	28.9	+9.8	+0.1	+0.0	+0.2	+0.0	39.0	49.4	-10.4	L2
34	2.395M	25.4	+9.8	+0.2	+0.1	+0.1	+0.0	35.6	46.0	-10.4	L2
35	2.429M	25.4	+9.8	+0.2	+0.1	+0.1	+0.0	35.6	46.0	-10.4	L2
36	261.262k	30.9	+9.8	+0.1	+0.0	+0.1	+0.0	40.9	51.4	-10.5	L2
37	528.874k	25.4	+9.8	+0.1	+0.0	+0.2	+0.0	35.5	46.0	-10.5	L2
38	3.242M	25.3	+9.8	+0.2	+0.1	+0.1	+0.0	35.5	46.0	-10.5	L2
39	307.076k	29.4	+9.8	+0.1	+0.0	+0.1	+0.0	39.4	50.0	-10.6	L2
40	2.221M	25.3	+9.8	+0.1	+0.1	+0.1	+0.0	35.4	46.0	-10.6	L2
41	3.182M	25.1	+9.8	+0.2	+0.1	+0.1	+0.0	35.3	46.0	-10.7	L2
42	4.080M	25.1	+9.8	+0.2	+0.1	+0.1	+0.0	35.3	46.0	-10.7	L2
43	305.622k	29.3	+9.8	+0.1	+0.0	+0.1	+0.0	39.3	50.1	-10.8	L2
44	357.980k	27.9	+9.8	+0.1	+0.0	+0.2	+0.0	38.0	48.8	-10.8	L2

45	2.812M	25.0	+9.8	+0.2	+0.1	+0.1	+0.0	35.2	46.0	-10.8	L2
46	279.442k	29.9	+9.8	+0.1	+0.0	+0.1	+0.0	39.9	50.8	-10.9	L2
47	327.438k	28.5	+9.8	+0.1	+0.0	+0.2	+0.0	38.6	49.5	-10.9	L2
48	328.892k	28.5	+9.8	+0.1	+0.0	+0.2	+0.0	38.6	49.5	-10.9	L2
49	498.332k Ave	21.4	+9.8	+0.1	+0.0	+0.2	+0.0	31.5	46.0	-14.5	L2
^	498.331k	37.2	+9.8	+0.1	+0.0	+0.2	+0.0	47.3	46.0	+1.3	L2
^	500.513k	36.2	+9.8	+0.1	+0.0	+0.2	+0.0	46.3	46.0	+0.3	L2

Test Setup Photos





Telecom Conducted Emissions

Test Notes: Conducted Disturbances at Mains Terminals, T-LISN method.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	D. Nguyen
Test Method:	EN 301 489-1	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares the EUT does not have telecommunication ports.		
Configuration:	NA		

Radiated Emissions

Test Notes: Radiated disturbances emanating from enclosure.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	D. Nguyen
Test Method:	EN 301 489-1	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares radiated emissions test is not applicable per EN 301 489-1 §8.2; this equipment either does not have associated ancillary equipment or the radio in combination with the ancillary equipment was tested under the relevant product standard for the effective use of spectrum.		
Configuration:	NA		

Harmonic Emissions

Test Notes: Conducted disturbances of current harmonics emanating from equipment power supply to low-voltage public power networks.

Test Setup/Conditions			
Test Location:	Brea	Test Engineer:	D. Nguyen
Test Method:	EN 61000-3-2	Test Date(s):	5/18/2017
Test Level:	Harmonic Emissions		
Declarations:	None		
Configuration:	1		

Test Equipment					
Asset #	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
02271	AC Power Source	California Instruments	1251WP	5/13/2016	5/13/2018
2890	Flicker & Harmonics Meter	EMC Partner AG	HAR1000-1P	5/29/2015	5/29/2017

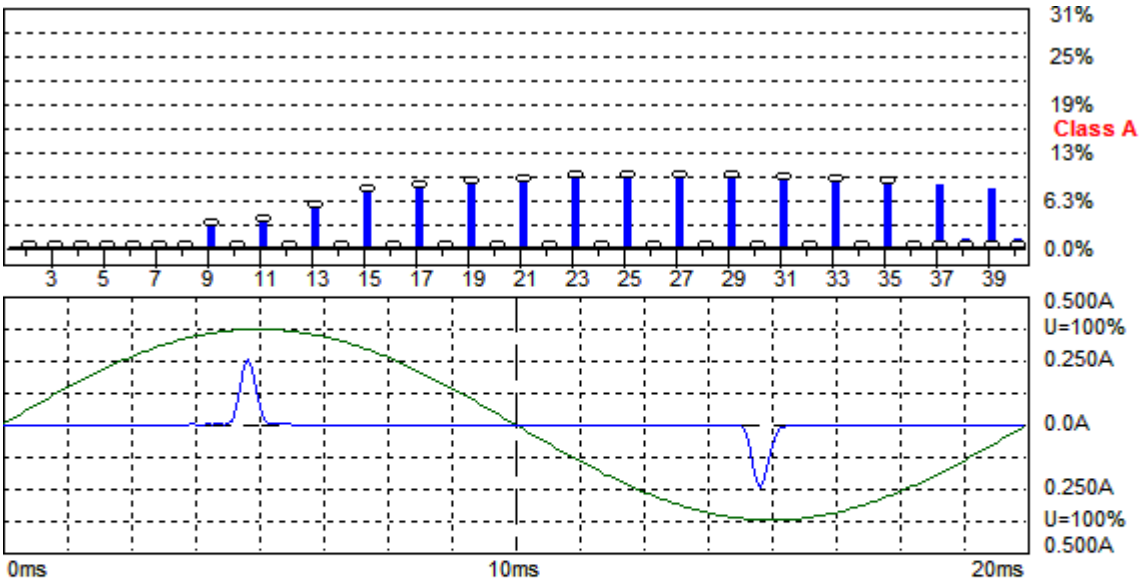
Harmonics Emissions Results Table		
Standard	**Test**	**Pass/Fail**
EN 61000-3-2	Harmonic Emissions	Pass

Test Data

Harmonic Emission - EN 61000-3-2

SMARTFIT Inc. WO. 99820

Operator Don Nguyen
Unit Configuration 1
Remarks : Normal operation mode



Harmonic Emission - IEC 61000-3-2 , EN 61000-3-2 , (EN60555-2) 5/18/2017 3:23:42 PM

Urms =	230.1	V	P =	3.411	W	THC =	0.041	A
Irms =	0.041	A	pf =	0.361				

Range: 0.5 A
V-nom: 230 V
TestTime: 5 min (100%)

Configuration 1 **Test completed, Result: PASSED**

Normal operation mode

HAR-1000 EMC-Partner

Full Bar : Actual Values
Empty Bar : Maximum Values
Blue : Current , Green : Voltage , Red : Failed

Measurement

SMARTFIT Inc. WO. 99820
Date : 5/18/2017 3:23:42 PM V4.21

Operator Don Nguyen
Unit Configuration 1
Remarks : Normal operation mode

Urms = 230.1V Freq = 50.065 Range: 0.5 A
Irms = 0.041A Ipk = 0.259A cf = 6.310
P = 3.411W S = 9.438VA pf = 0.361
THDi = 278 % THDu = 0.10 % Class A

Test - Time : 5min (100 %)

Test completed, Result: PASSED

Order	Freq. [Hz]	Iavg [A]	Irms [A]	Imax [A]	Limit [A]	Status
1	50	0.0148	0.0148	0.0149		
2	100	0.0000	0.0001	0.0001	1.0800	
3	150	0.0125	0.0125	0.0126	2.3000	
4	200	0.0000	0.0001	0.0001	0.4300	
5	250	0.0124	0.0124	0.0125	1.1400	
6	300	0.0000	0.0001	0.0001	0.3000	
7	350	0.0123	0.0123	0.0123	0.7700	
8	400	0.0000	0.0001	0.0002	0.2300	
9	450	0.0120	0.0120	0.0121	0.4000	
10	500	0.0000	0.0001	0.0002	0.1840	
11	550	0.0117	0.0117	0.0118	0.3300	
12	600	0.0000	0.0002	0.0002	0.1533	
13	650	0.0114	0.0114	0.0114	0.2100	
14	700	0.0000	0.0002	0.0002	0.1314	
15	750	0.0110	0.0110	0.0110	0.1500	
16	800	0.0000	0.0002	0.0002	0.1150	
17	850	0.0105	0.0105	0.0106	0.1324	
18	900	0.0000	0.0002	0.0002	0.1022	
19	950	0.0100	0.0100	0.0100	0.1184	
20	1000	0.0000	0.0002	0.0003	0.0920	
21	1050	0.0095	0.0095	0.0095	0.1071	
22	1100	0.0000	0.0002	0.0003	0.0836	
23	1150	0.0089	0.0089	0.0089	0.0978	
24	1200	0.0000	0.0002	0.0003	0.0767	
25	1250	0.0083	0.0083	0.0084	0.0900	
26	1300	0.0000	0.0003	0.0003	0.0708	
27	1350	0.0077	0.0078	0.0078	0.0833	
28	1400	0.0000	0.0003	0.0003	0.0657	
29	1450	0.0071	0.0072	0.0072	0.0776	
30	1500	0.0000	0.0003	0.0003	0.0613	
31	1550	0.0065	0.0066	0.0066	0.0726	
32	1600	0.0000	0.0003	0.0004	0.0575	
33	1650	0.0059	0.0060	0.0060	0.0682	
34	1700	0.0000	0.0003	0.0004	0.0541	

35	1750	0.0053	0.0054	0.0054	0.0643
36	1800	0.0000	0.0003	0.0004	0.0511
37	1850	0.0000	0.0048	0.0048	0.0608
38	1900	0.0000	0.0003	0.0004	0.0484
39	1950	0.0000	0.0042	0.0042	0.0577
40	2000	0.0000	0.0003	0.0004	0.0460

Test Setup Photo



Voltage Fluctuation and Flicker Emissions

Test Notes: Conducted disturbances of voltage fluctuations and flicker emanating from equipment power supply to low-voltage public power networks.

Test Setup/Conditions			
Test Location:	Brea	Test Engineer:	D. Nguyen
Test Method:	EN 61000-3-3	Test Date(s):	5/18/2017
Test Level:	Flicker Emissions		
Declarations:	None		
Configuration:	1		

Test Equipment					
Asset #	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
02271	AC Power Source	California Instruments	1251WP	5/13/2016	5/13/2018
2890	Flicker & Harmonics Meter	EMC Partner AG	HAR1000-1P	5/29/2015	5/29/2017

Voltage Fluctuation and Flicker Emissions Results Table		
Standard	**Test**	**Pass/Fail**
EN 61000-3-3	Voltage Fluctuations and Flicker Emissions	Pass

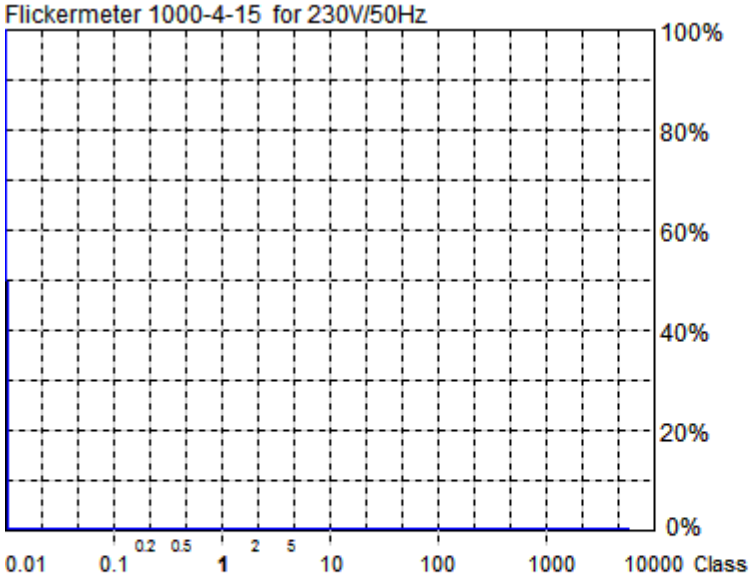
Test Data

Flicker Emission EN 61000-3-3

SMARTFIT Inc. WO. 99820

Operator Don Nguyen
Unit Configuration 1
Remarks : Normal operation mode

Flickermeter 1000-4-15 for 230V/50Hz



Actual Flicker (Fli):	**0.00**
Short-term Flicker (Pst):	**0.07**
Limit (Pst):	1.00
Long-term Flicker (Plt):	**0.07**
Limit (Plt):	0.65
Maximum Relative Volt. Change (dmax):	**0.00%**
Limit (dmax):	4.00%
Relative Steady-state Voltage Change (dc):	**0.00%**
Limit (dc):	3.00%
Maximum Interval exceeding 3.00% (dt):	**0.00ms**
Limit (dt>Lim):	500ms

Flicker Emission - IEC 61000-3-3 , EN 61000-3-3

5/18/2017 3:34:57 PM

Urms = 230.1 V P = 3.399 W
Irms = 0.039 A pf = 0.383

Range: 0.5 A
V-nom: 230 V
TestTime: 10 min (100%)

Configuration 1

Test completed, Result: PASSED

Normal operation mode

HAR-1000 EMC-Partner

Full Bar : Actual Values
Empty Bar : Maximum Values
Circles : Average Values
Blue : Current , Green : Voltage , Red : Failed

Measurement

```
SMARTFIT Inc. WO. 99820
Date : 5/18/2017 3:34:57 PM V4.21

Operator                        Don Nguyen
Unit                            Configuration 1
Remarks :                       Normal operation mode

Urms =  230.1V  Freq =  50.065  Range:  0.5 A
Irms =  0.039A  Ipk  =  0.225A  cf   =  5.842
P    =  3.399W  S    =  8.876VA pf   =  0.383

Test - Time :   1 x 10min = 10min      ( 100 %)

LIN (Line Impedance Network) :  L: 0.24ohm +j0.15ohm N: 0.16ohm +j0.10ohm

Limits :        Plt  :  0.65    Pst   : 1.00
                dmax :  4.00 %  dc    : 3.00 %
                dtLim:  3.00 %  dt>Lim: 500ms

Test completed, Result: PASSED

        dmax
        [%]
1       0.000
```

Test Setup Photo



Immunity

Immunity Performance Criteria Definitions

Designation	Definition
Standard / Mfg.	**Standard Requirement / Manufacturer's Declaration**
A	**During test:** Operate as intended No loss of function No unintentional responses **After test:** Operate as intended No loss of function No degradation of performance No loss of stored data or user programmable functions **Performance criterion A applies for immunity tests with phenomena of a continuous nature.**
B	**During test:** May show loss of function No unintentional responses **After test:** Operate as intended Lost function(s) shall be self-recoverable No degradation of performance No loss of stored data or user programmable functions **Performance criterion B applies for immunity tests with phenomena of a transient nature.**
Per Standard	For a voltage interruption the following performance criteria apply: • in the case where the equipment is fitted with or connected to a battery back-up, the performance criteria for transient phenomena for transmitters or for receivers shall apply (see clause 6 of the relevant part of ETSI EN 301 489 series [i.13] dealing with the particular type of radio equipment); • in the case where the equipment is powered solely from the AC mains supply (without the use of a parallel battery back-up) volatile user data may have been lost and if applicable the communication link need not to be maintained and lost functions should be recoverable by user or operator; • no unintentional responses shall occur at the end of the test; • in the event of loss of function(s) or in the event of loss of user stored data, this fact shall be recorded in the test report; • for ancillary equipment the pass/failure criteria supplied by the manufacturer (see clause 6.4) shall apply, unless the ancillary equipment is tested in connection with a receiver or transmitter in which case the corresponding performance criteria above shall apply.
CT	Continuous [phenomena applied to] Transmitters
CR	Continuous [phenomena applied to] Receivers
TT	Transient [phenomena applied to] Transmitters
TR	Transient [phenomena applied to] Receivers

Immunity Setup and Monitoring

Equipment Setup: The EUT is placed on table top. The EUT is connected to 6Vdc adapter. The input voltage to ACDC adapter is 230Vac/50Hz. A 3.7Vdc lithium battery is installed inside the EUT.
The EUT is tested in preferred orientation declared by the manufacturer.

TX mode: The EUT is paired with a support CPU Controller and exercise intended functionalities.
Operating frequency: 434MHz
Antenna gain: 0dBi
Software output power: <1mW

Standby mode: The EUT is set in standby mode. Probe antenna is placed next the EUT and connected to a spectrum analyzer to monitor any unintended transmission.

Degradation Criteria: Loss of communication between the EUT and the support equipment or any abnormal behavior is constituted as failure.

Monitoring: Visual inspection of LED activities and monitoring the sound from built-in speakers of the support CPU Controller.

Electrostatic Discharge

Test Notes: Immunity threat to simulate static electrical charges directly from an operator or from adjacent objects.

Test Setup/Conditions			
Test Location:	Brea Lab B	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-2	Test Date(s):	5/24/2017
Test Level:	Contact: +/- 4kV Air: +/- 2, +/-4 and +/-8kV		
Declarations:	None		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	22	Relative Humidity (%):	45	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
01650	ESD Simulator	Keytek	MZ-15/ec	10/13/2016	10/13/2017

ESD Contact Discharge Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
HCP	No degradation	
VCP	No degradation	
Front	No degradation	
Back	No degradation	
Left Side	No degradation	
Right Side	No degradation	
Top	No degradation	
Bottom	No degradation	

ESD Air Discharge Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
Front	No degradation	
Back	No degradation	
Left Side	No degradation	
Right Side	No degradation	
Top	No degradation	
Bottom	No degradation	

ESD Contact Discharge Results		
Mode of operation:	Standby	
Surface Tested	**Test Results**	**Test Observations**
HCP	No degradation	No unintended transmission occurred.
VCP	No degradation	No unintended transmission occurred.
Front	No degradation	No unintended transmission occurred.
Back	No degradation	No unintended transmission occurred.
Left Side	No degradation	No unintended transmission occurred.
Right Side	No degradation	No unintended transmission occurred.
Top	No degradation	No unintended transmission occurred.
Bottom	No degradation	No unintended transmission occurred.

ESD Air Discharge Results		
Mode of operation:	Standby	
Surface Tested	**Test Results**	**Test Observations**
Front	No degradation	No unintended transmission occurred.
Back	No degradation	No unintended transmission occurred.
Left Side	No degradation	No unintended transmission occurred.
Right Side	No degradation	No unintended transmission occurred.
Top	No degradation	No unintended transmission occurred.
Bottom	No degradation	No unintended transmission occurred.

ESD Test Summary		
Standard Minimum Performance Criteria:	TT/TR/B	
For this test, the EUT met performance criteria:	TT/TR/A	Result: Pass

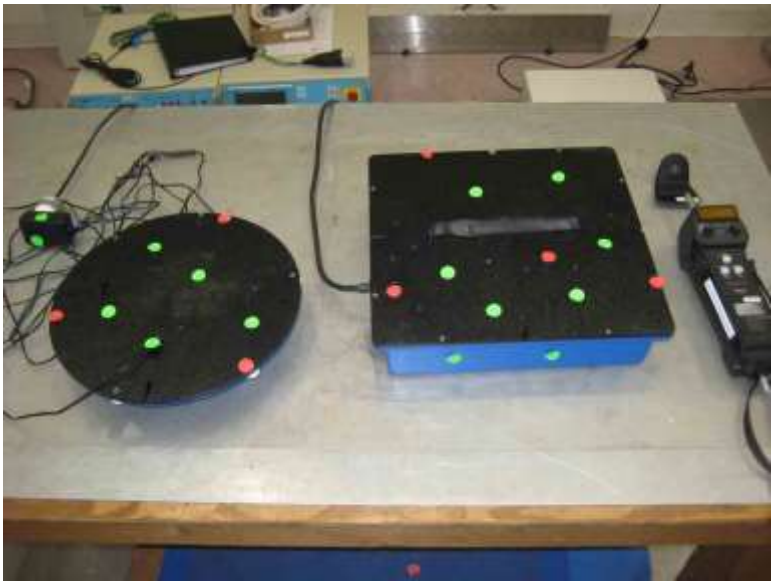
Test Setup Photos

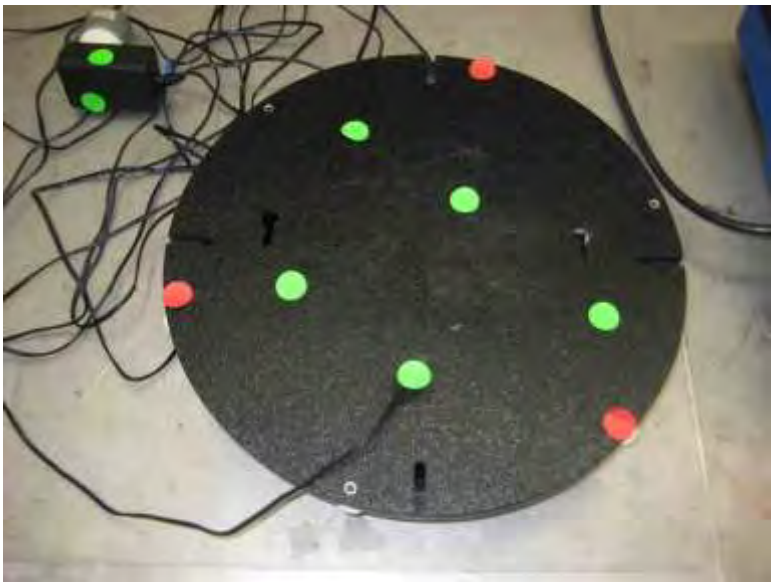
Red = Contact Discharge
Green = Air Discharge











Radiated Immunity

Test Notes: Immunity threat to simulate radiated electro-magnetic fields.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-3	Test Date(s):	5/19/2017
Test Level:	80MHz to 6000MHz (except exclusion band), 3V/m @ 80% AM 1kHz		
Declarations:	None		
Configuration:	A		

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
BC 61000-4-3	Chamber Cal	CKC	Brea Chamber	1/4/2017	1/4/2019
03354	Log Periodic Antenna	Amplifier Research	ATR80M6G	1/4/2017	1/4/2019
00627	Amplifier	AR	50W1000A	1/6/2017	1/6/2019
01612	Directional Coupler	Werlatone	C2630	7/6/2015	7/6/2017
02869	Spectrum Analyzer	Agilent	E4440A	7/8/2016	7/8/2017
00636	Field Monitor	AR	FM2000	10/22/2015	10/22/2017
00638	Field Probe	AR	FP2000	10/22/2015	10/22/2017
P05301	Directional Coupler	ATM	C223E-20	5/6/2016	5/6/2018
02688	Amplifier	Ophir	5161F	9/22/2016	9/22/2018
P07070	Cable	Andrew	LDF1-50-55	1/4/2017	1/4/2019
P07071	Cable	Andrew	LDF1-50-55	1/4/2017	1/4/2019
P07072	Cable	Andrew	LDF1-50-6	1/4/2017	1/4/2019
P06663	Cable	Huber & Suhner Inc.	Sucoflex 104A	4/5/2016	4/5/2018
03420	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P03363	Attenuator	ATM	CH225H-35	7/1/2015	7/1/2017
02551	RF Generator	HP	8673M	5/20/2015	5/20/2017
00838	Arbitrary Waveform Generator	HP	33120A	1/25/2017	1/25/2019
03596	Amplifier	Amplifier Research	30S1G6	3/3/2017	3/3/2019

Radiated Immunity Results		
Mode of operation:	Normal Operation	
Surface Tested	**Test Results**	**Test Observations**
Front	No degradation	
Back	No degradation	
Left Side	No degradation	
Right Side	No degradation	

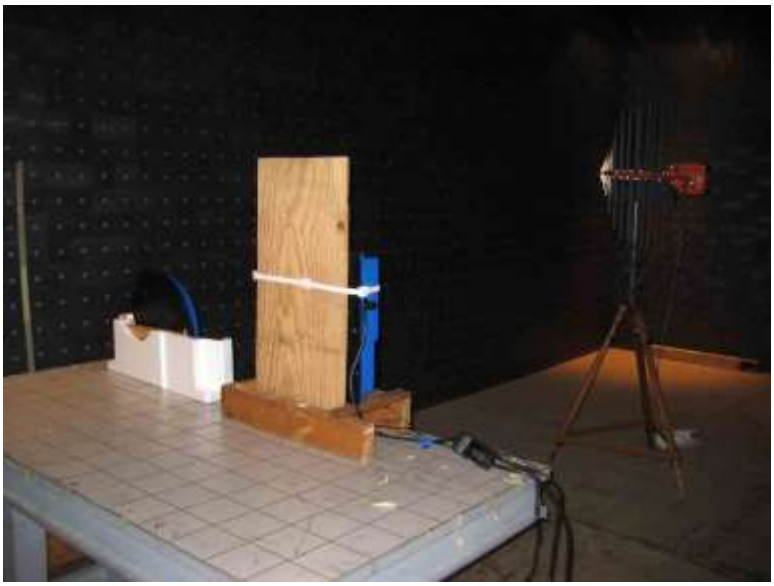
Radiated Immunity Results		
Mode of operation:	Standby	
Surface Tested	**Test Results**	**Test Observations**
Front	No degradation	
Back	No degradation	
Left Side	No degradation	
Right Side	No degradation	

Frequency Range	Test Distance (m)
80-6000MHz	1.75

Radiated Immunity Test Summary		
Standard Minimum Performance Criteria:	CT/CR/A	
For this test, the EUT met performance criteria:	CT/CR/A	Result: Pass

Test Setup Photos





Electrical Fast Transients / Bursts – AC/DC Mains Power Lines

Test Notes: Immunity threat to simulate electrical fast transient signals and bursts.

Test Setup/Conditions			
Test Location:	Brea Lab B	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-4	Test Date(s):	5/24/2017
Test Level:	AC Mains: 1kV, 5kHz PRF DC Mains: NA		
Declarations:	DC Mains: The manufacturer declares that the EUT does not operate from DC mains.		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	22	Relative Humidity (%):	45	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00844	Simulator ESD / Surge / EFT / Voltage Dip	EM Test	USC 500M4	8/9/2016	8/9/2017

EFT/B AC Mains Power Lines Results Table			
Mode of operation:	Normal Operation		
Coupling Mode		**Test Results**	**Test Observations**
Line to Ground		No degradation	
Neutral to Ground		No degradation	
Protective Earth to Ground		No degradation	
Line/Neutral/Protective Earth to Ground.		No degradation	

EFT/B AC Mains Power Lines Results Table			
Mode of operation:	Standby		
Coupling Mode		**Test Results**	**Test Observations**
Line to Ground		No degradation	No unintended transmission occurred.
Neutral to Ground		No degradation	No unintended transmission occurred.
Protective Earth to Ground		No degradation	No unintended transmission occurred.
Line/Neutral/Protective Earth to Ground.		No degradation	No unintended transmission occurred.

EFT/B Test Summary		
Standard Minimum Performance Criteria:	TT/TR/B	
For this test, the EUT met performance criteria:	TT/TR/A	Result: Pass

Test Setup Photo



Electrical Fast Transients / Bursts - IO Lines

Test Notes: Immunity threat to simulate electrical fast transient signals and bursts.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	D. Nguyen
Test Method:	EN 61000-4-4	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares that the EUT has no IO port accessible to user.		
Configuration:	NA		

Surge

Test Notes: Immunity threat to simulate unidirectional surges caused by over voltages from switching and lightning transients.

Test Setup/Conditions			
Test Location:	Brea Lab B	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-5	Test Date(s):	5/24/2017
Test Level:	AC Mains: 1kV Line to Line , 2kV Line to Ground		
Declarations:	None		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	22	Relative Humidity (%):	45	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00844	Simulator ESD / Surge / EFT / Voltage Dip	EM Test	USC 500M4	8/9/2016	8/9/2017

Surge AC Mains Results		
Mode of operation:	Normal Operation	
Coupling Mode	**Test Results**	**Test Observations**
Line-to-Line	No degradation	
Line-to-Ground	No degradation	
Neutral-to-Ground	No degradation	

Surge AC Mains Results		
Mode of operation:	Standby	
Coupling Mode	**Test Results**	**Test Observations**
Line-to-Line	No degradation	No unintended transmission occurred.
Line-to-Ground	No degradation	No unintended transmission occurred.
Neutral-to-Ground	No degradation	No unintended transmission occurred.

Surge Test Summary		
Standard Minimum Performance Criteria:	TT/TR/B	
For this test, the EUT met performance criteria:	TT/TR/A	Result: Pass

Test Setup Photo



Surge IO Ports

Test Notes: Immunity threat to simulate unidirectional surges caused by over voltages from switching and lightning transients.

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	D. Nguyen
Test Method:	NA	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares that the EUT has no IO port accessible to user.		
Configuration:	NA		

Conducted Immunity

Test Notes: Immunity threat to simulate low frequency radiated electro-magnetic fields having at least one conducting cable.

Test Setup/Conditions			
Test Location:	Brea Lab C	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-6	Test Date(s):	5/22/2017
Test Level:	AC Mains: 150kHz to 80MHz (except exclusion band), 3 Vrms @ 80% AM 1kHz DC Mains: NA IO Lines: NA		
Declarations:	DC Mains: The manufacturer declares that the EUT does not operate from DC mains. IO Lines: The manufacturer declares that the EUT has no IO port accessible to user.		
Configuration:	A		

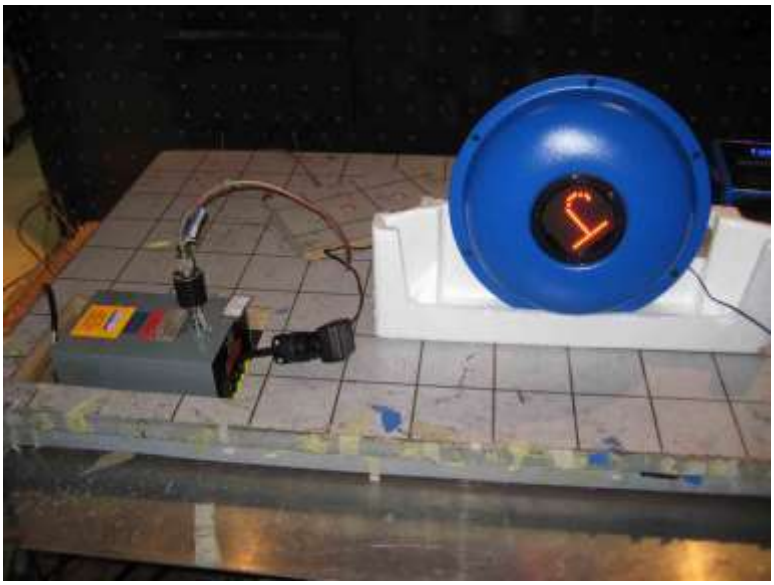
Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
03049	Coupling Decoupling Network	FCC	FCC-801-M3-25	11/28/2016	11/28/2017
00628	Amplifier	AR	100A100	1/6/2017	1/6/2019
02869	Spectrum Analyzer	Agilent	E4440A	7/8/2016	7/8/2017
P04357	Coaxial Cable	Carolina Wire	RG142	4/4/2016	4/4/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017

Conducted Immunity AC Mains Results			
Mode of operation:	Normal Operation		
Port Tested		**Test Results**	**Test Observations**
AC Mains		No degradation	

Conducted Immunity AC Mains Results			
Mode of operation:	Standby		
Port Tested		**Test Results**	**Test Observations**
AC Mains		No degradation	

Conducted Immunity Test Summary		
Standard Minimum Performance Criteria:	CT/CR/A	
For this test, the EUT met performance criteria:	CT/CR/A	Result: Pass

Test Setup Photo



Voltage Dips and Interrupts

Test Notes: Immunity threat to simulate low-voltage power supply network voltage dips and interruptions of long duration.

Test Setup/Conditions			
Test Location:	Brea Lab B	Test Engineer:	Don Nguyen
Test Method:	EN 61000-4-11	Test Date(s):	5/24/2017
Test Level:	0% residual for 0.5 cycle, 0% residual for 1 cycle, 70% residual for 25 cycles and 0% residual for 250 cycles		
Declarations:	None		
Configuration:	A		

Environmental Conditions					
Temperature (ºC)	22	Relative Humidity (%):	45	Atmospheric Pressure (kPa):	100

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
01379	Variac	Superior Electric	1256D	8/19/2016	8/19/2017
00844	Simulator ESD / Surge / EFT / Voltage Dip	EM Test	USC 500M4	8/9/2016	8/9/2017

Voltage Dips and Interruptions Results					
Mode of operation:	Normal Operation				
Ux (%)	**Duration (ms)**	**Minimum Criteria**	**Test Results**	**EUT Meets Criteria**	**Test Observations**
0	10	TT/TR/B	No degradation	TT/TR/A	
0	20	TT/TR/B	No degradation	TT/TR/A	
70	500	TT/TR/B	No degradation	TT/TR/A	
0	5000	Per Standard	No degradation	Per Standard	

Voltage Dips and Interruptions Results					
Mode of operation:	Standby				
Ux (%)	**Duration (ms)**	**Minimum Criteria**	**Test Results**	**EUT Meets Criteria**	**Test Observations**
0	10	TT/TR/B	No degradation	TT/TR/A	No unintended transmission occurred
0	20	TT/TR/B	No degradation	TT/TR/A	No unintended transmission occurred
70	500	TT/TR/B	No degradation	TT/TR/A	No unintended transmission occurred
0	5000	Per Standard	No degradation	Per Standard	No unintended transmission occurred

Voltage Dips and Interruptions Test Summary	
With combination of criteria listed above, the resultant overall assessment is:	Pass

Test Setup Photo



Transients and Surges in the Vehicle Environment – 12/24VDC Supply

Test Setup/Conditions			
Test Location:	NA	Test Engineer:	D. Nguyen
Test Method:	ISO 7637-2	Test Date(s):	NA
Test Level:	NA		
Declarations:	The manufacturer declares that the EUT does not operate from 12 or 24 VDC vehicle supply.		
Configuration:	NA		

SUPPLEMENTAL INFORMATION

Measurement Uncertainty

Uncertainty Value	Parameter
4.73 dB	Radiated Emissions
3.34 dB	Mains Conducted Emissions
3.30 dB	Disturbance Power

Uncertainties reported are worst case for all CKC Laboratories' sites and represent expanded uncertainties expressed at approximately the 95% confidence level using a coverage factor of k=2.

Emissions Test Details

TESTING PARAMETERS

Unless otherwise indicated, the following configuration parameters are used for equipment setup: The cables were routed consistent with the typical application by varying the configuration of the test sample. Interface cables were connected to the available ports of the test unit. The effect of varying the position of the cables was investigated to find the configuration that produced maximum emissions. Cables were of the type and length specified in the individual requirements. The length of cable that produced maximum emissions was selected.

The equipment under test (EUT) was set up in a manner that represented its normal use, as shown in the setup photographs. Any special conditions required for the EUT to operate normally are identified in the comments that accompany the emissions tables.

The emissions data was taken with a spectrum analyzer or receiver. Incorporating the applicable correction factors for distance, antenna, cable loss and amplifier gain, the data was reduced as shown in the table below. The corrected data was then compared to the applicable emission limits. Preliminary and final measurements were taken in order to ensure that all emissions from the EUT were found and maximized.

CORRECTION FACTORS

The basic spectrum analyzer reading was converted using correction factors as shown in the highest emissions readings in the tables. For radiated emissions in dBµV/m, the spectrum analyzer reading in dBµV was corrected by using the following formula. This reading was then compared to the applicable specification limit. Individual measurements were compared with the displayed limit value in the margin column. The margin was calculated based on subtracting the limit value from the corrected measurement value; a positive margin represents a measurement exceeding the limit, while a negative margin represents a measurement less than the limit.

SAMPLE CALCULATIONS		
	Meter reading	(dBµV)
+	Antenna Factor	(dB/m)
+	Cable Loss	(dB)
-	Distance Correction	(dB)
-	Preamplifier Gain	(dB)
=	Corrected Reading	(dBµV/m)

TEST INSTRUMENTATION AND ANALYZER SETTINGS

The test instrumentation and equipment listed were used to collect the emissions data. A spectrum analyzer or receiver was used for all measurements. Unless otherwise specified, the following table shows the measuring equipment bandwidth settings that were used in designated frequency bands. For testing emissions, an appropriate reference level and a vertical scale size of 10 dB per division were used.

MEASURING EQUIPMENT BANDWIDTH SETTINGS PER FREQUENCY RANGE			
TEST	BEGINNING FREQUENCY	ENDING FREQUENCY	BANDWIDTH SETTING
CONDUCTED EMISSIONS	150 kHz	30 MHz	9 kHz
RADIATED EMISSIONS	30 MHz	1000 MHz	120 kHz
RADIATED EMISSIONS	1000 MHz	>1 GHz	1 MHz

SPECTRUM ANALYZER/RECEIVER DETECTOR FUNCTIONS

The notes that accompany the measurements contained in the emissions tables indicate the type of detector function used to obtain the given readings. Unless otherwise noted, all readings were made in the "positive peak" detector mode. Whenever a "quasi-peak" or "average" reading was recorded, the measurement was annotated with a "QP" or an "Ave" on the appropriate rows of the data sheets. In cases where quasi-peak or average limits were employed and data exists for multiple measurement types for the same frequency then the peak measurement was retained in the report for reference, however the numbering for the affected row was removed and an arrow or caret ("^") was placed in the far left-hand column indicating that the row above takes precedence for comparison to the limit. The following paragraphs describe in more detail the detector functions and when they were used to obtain the emissions data.

Peak

In this mode, the spectrum analyzer or receiver recorded all emissions at their peak value as the frequency band selected was scanned. By combining this function with another feature called "peak hold," the measurement device had the ability to measure intermittent or low duty cycle transient emission peak levels. In this mode the measuring device made a slow scan across the frequency band selected and measured the peak emission value found at each frequency across the band.

Quasi-Peak

Quasi-peak measurements were taken using the quasi-peak detector when the true peak values exceeded or were within 2 dB of a quasi-peak specification limit. Additional QP measurements may have been taken at the discretion of the operator.

Average

Average measurements were taken using the average detector when the true peak values exceeded or were within 2 dB of an average specification limit. Additional average measurements may have been taken at the discretion of the operator. If the specification or test procedure requires trace averaging, then the averaging was performed using 100 samples or as required by the specification. All other average measurements are performed using video bandwidth averaging. To make these measurements, the test engineer reduces the video bandwidth on the measuring device until the modulation of the signal is filtered out. At this point the measuring device is set into the linear mode and the scan time is reduced.

Immunity Test Details

Radiated Immunity Field Intensity at 0.4m

Chamber (80-1000)	1V/m Testing	3V/m Testing	10V/m Testing	20V/m Testing	30V/m Testing
Fremont C1	1.5	5.2	11.0	NA	NA
Fremont C2	1.9	2.7	7.1	16.2	NA
Mariposa	1.0	3.3	9.9	NA	NA
Brea	0.6	2.2	12.4	NA	NA
Bothell C1	1.7	3.1	9.0	17.0	25.4
Bothell C2	0.6	1.9	6.8	NA	NA

NA = Not applicable because these levels are not performed in this chamber.